

A Comparative Analysis of Few-Shot and Per-Scene NeRF models on Sparse-View Datasets

Abstract—This work compares Generalizable NeRFs and Fast Optimization Per-Scene trained models applied to sparse-view datasets, evaluating the impact of these feature extraction architectures on novel view synthesis performance. Through a quantitative analysis using visual fidelity (PSNR, SSIM, LPIPS) and computational efficiency metrics (convergence time, VRAM footprint, inference time), this study highlights the trade-offs between the global generalization of Transformers and the local processing efficiency of convolutions, as well as the critical impact of spatial scale mismatch in per-scene optimized models. Ultimately, it aims to guide the selection of 3D rendering models for practical scenarios often constrained by hardware limitations and data scarcity, meanwhile

I. INTRODUCTION

Neural Radiance Fields (NeRFs) have consolidated themselves in recent years as the state-of-the-art for novel view synthesis, in both industry and academia [1]–[3]. However, the most advanced neural rendering techniques demand vast amounts of: (i) computational power; and (ii) multiple views in the training dataset. Both challenges make practical applications—such as mobile scanning, virtual try-ons, and synthesis in reduced datasets—unfeasible. To address this problem, projects such as Instant-NGP (*Instant Neural Graphics Primitives*), Nerfacto, PixelNeRF, and GNT (*Generalizable NeRF Transformer*) [2], [4]–[6], among others, attempt to expand the state-of-the-art and, consequently, tackle analogous themes, such as: (i) spatial parameter *encoding* [4]; (ii) the use of hybrid architectures combining Convolutional Neural Networks (CNNs) and Vision Transformers (ViT) [6]–[8]; and (iii) the pre-computation (*baking*) of models into structures with a smaller memory footprint [9].

Consequently, this work aims to explore the use of different neural radiance field models and their respective optimization techniques by applying them to sparse-view datasets, promoting an exploratory analysis between computational performance and the visual quality of the generated volumes. To evaluate reconstruction fidelity, visual quality metrics widely established in the literature will be used, such as PSNR (*Peak Signal-to-Noise Ratio*), SSIM (*Structural Similarity Index Measure*) [10], and LPIPS (*Learned Perceptual Image Patch Similarity*) [11]. In parallel, the quantification of operational costs will be based on systemic performance metrics, including convergence time during training, inference speed, and memory *footprint*, especially VRAM (*Video Random Access Memory*) consumption. Although works like [12] and [13] have evaluated NeRFs in limited data regimes, and projects like NerfBaselines [14] have reported results on traditional and per-scene training-based NeRFs, a systematic comparison

between generalizable models with convolutional extractors versus attention-based ones, under controlled hardware conditions and reproducible splits, remains unexplored in the literature. In short, this investigation provides a consolidated panorama of the *trade-offs* between computational efficiency and visual fidelity, establishing clear guidelines for the viability and practical application of novel view synthesis models in highly constrained scenarios, particularly regarding data and hardware limitations.

Given this problem, the contributions of this work are: (i) a systematic and reproducible evaluation of few-shot and per-scene NeRF models under three data regimes; and (ii) the public release of integration wrappers for PixelNeRF and GNT into the Nerfstudio framework at <https://github.com/felipe-gsilva/few-shot-nerfs>, adapting their input and output interfaces to comply with the framework’s unified evaluation protocol, without modifying the pre-trained weights or original inference logic, to allow reproducible comparisons under controlled conditions.

II. MATERIALS AND METHODS

This section outlines the methodology adopted for the experimental phase of this work. Figure 1 illustrates the proposed execution protocol and details the isolated environment configuration. To systematically evaluate the trade-offs between visual fidelity and computational efficiency, the subsequent steps of this study are organized as follows: (i) the selection and sampling strategy of the dataset; (ii) the chosen neural rendering architectures and their respective training regimes; and (iii) the specific criteria and metrics used for evaluation.

A. Dataset & Splits

The experiments were conducted on the LLFF (*Local Light Field Fusion*) dataset [15], which comprises 8 real-world scenes (*fern, flower, fortress, horns, leaves, orchids, room, trex*) captured with a forward-facing moving camera. Camera poses and depth bounds were obtained via COLMAP, following standard dataset procedures. Furthermore, images were evaluated across different resolutions, specifically downscale 4 (1008×756) and downscale 8 (504×378).

The complete data splitting protocol follows the few-shot evaluation configuration established by Niemeyer et al. [16]. Specifically, the testing split was generated by reserving images with indices that are multiples of 8 (`11ffhold=8`), while the training views for the sparse regimes were strictly limited to $N \in \{3, 6, 10\}$. The selection of these training views was performed using two distinct strategies: (i) *uniform*, where

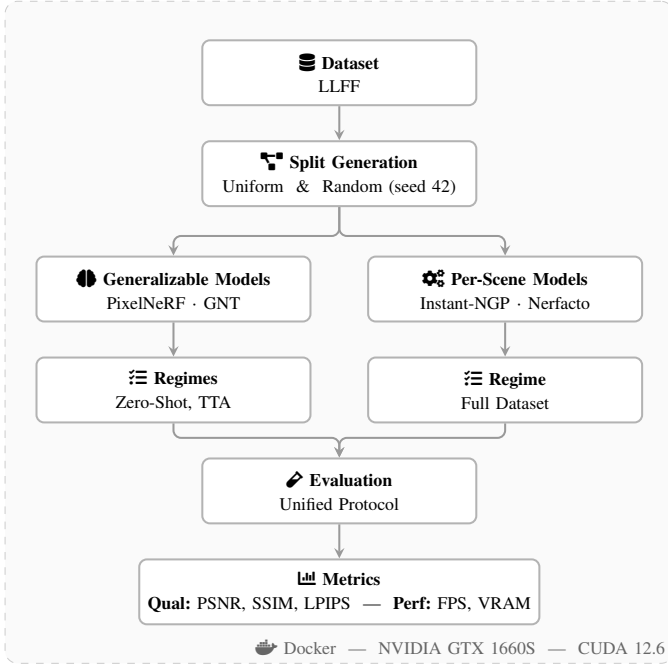


Fig. 1. Proposed methodology pipeline. The flowchart details the progression from data preparation to the bifurcated training paradigms, culminating in a unified evaluation protocol. The entire pipeline executes within an isolated containerized environment.

TABLE I
IMPACT OF THE TRAINING VIEW SELECTION STRATEGY (*uniform* vs. *random*) ON AVERAGE VISUAL QUALITY. EVALUATED UNDER THE TTA REGIME WITH $N = 10$ VIEWS ON THE *fern* SCENE.

Model	Split	PSNR (dB) $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	Δ PSNR
PixelNeRF	Uniform	6.98	0.047	0.846	+0.54
	Random	6.44	0.042	0.840	
GNT	Uniform	10.34	0.179	0.738	+0.15
	Random	10.19	0.184	0.742	
Instant-NGP	Uniform	13.57	0.231	0.627	+0.06
	Random	13.51	0.229	0.630	
Nerfacto	Uniform	12.04	0.227	0.809	+0.11
	Random	11.93	0.226	0.865	

views are sampled via linear interpolation over the dataset indices, distributing them evenly along the camera trajectory; and (ii) *random*, where views are randomly selected using a fixed seed (42) to ensure reproducibility. Comparing these two strategies allows us to evaluate the models’ sensitivity to the spatial distribution of the input views, as demonstrated in Table I. Given the empirical advantage of uniform sampling (+0.54 dB PSNR on the *fern* scene), this strategy was adopted for all subsequent experiments.

B. Models & Training Regimes

This study evaluates representative models for novel view synthesis under few-shot constraints, encompassing distinct architectural paradigms. The generalizable approaches selected are: (i) **PixelNeRF** [5], which utilizes Convolutional Neural Networks (CNNs) to extract spatial features and inject them

TABLE II
TRAINING AND OPTIMIZATION HYPERPARAMETERS ACROSS EVALUATED REGIMES. TTA CONFIGURATIONS DENOTE THE FINE-TUNING PHASE.

Model	Regime	Iterations (Steps)	Rays / Batch	Learning Rate (η)
PixelNeRF	TTA	[PASSOS]	[RAIOS]	10^{-3}
	Zero-Shot	0	[RAIOS]	N/A
GNT	TTA	[PASSOS]	[RAIOS]	10^{-3}
	Zero-Shot	0	[RAIOS]	N/A
Instant-NGP	Per-Scene	[PASSOS]	[RAIOS]	10^{-3}
Nerfacto	Per-Scene	[PASSOS]	[RAIOS]	10^{-3}

into the renderer via projection; and (ii) **Generalizable NeRF Transformer (GNT)** [6], which replaces both the feature extractor and the differentiable renderer with cross-attention modules. As comparative baselines, we evaluate two highly efficient *per-scene* architectures: **Instant-NGP** [4], which replaces traditional MLPs with a multiscale hash table encoding, and **Nerfacto** [2], an aggregate model combining state-of-the-art heuristics for rapid convergence and high fidelity.

The experimental execution was structured across three distinct regimes: (i) **Zero-Shot**: The ideal paradigm for extreme compute constraints, performing instant inference using pre-trained weights without any local optimization on the novel scene; (ii) **Test-Time Adaptation (TTA)**: An intermediate regime where the pre-trained generalizable models are fine-tuned using the sparse input views ($N \in \{3, 6, 10\}$), allowing rapid adaptation to geometric discrepancies with low computational overhead; (iii) **Per-Scene (100%)**: Representing the upper bound for visual fidelity, this regime optimizes the baseline models from scratch using the entirety of available images for a specific scene.

To ensure a strictly equitable evaluation protocol, all models were unified under the *Nerfstudio* framework [2]. Instant-NGP and Nerfacto are natively supported by the framework. Conversely, integration wrappers were developed for PixelNeRF and GNT, porting their original DataManager and Pipeline components to comply with Nerfstudio’s modular interface. This unified environment leverages *tiny-cuda-nn* [17] for GPU acceleration and standardizes the logging infrastructure.

For consistency across the evaluation, all models were optimized using Adam with a standard learning rate of $\eta = 10^{-3}$ and zero weight decay. While the original GNT and PixelNeRF literature suggests slightly lower learning rates (between 10^{-4} and 5×10^{-4}), adopting $\eta = 10^{-3}$ strictly standardizes the hyperparameter baseline across all architectures a criterion that factors into the subsequent TTA stability analysis, as can be seen in Table II. The complete wrapper integration code is publicly available at <https://github.com/Felipe-gsilva/Few-Shot-NeRFs/>. Additionally, the modular wrappers can be downloaded directly as packages from PyPI and piwheels (<https://www.piwheels.org/project/nerfstudio-gnt/> and <https://www.piwheels.org/project/nerfstudio-pixelnerf/>), while the reproducible experiment scripts and environment configurations are hosted at <https://github.com/Felipe-gsilva/nerf-ann-paper/>.

C. Evaluation & Metrics

The evaluation framework assesses three complementary dimensions: visual fidelity, computational efficiency, and memory consumption. Final metrics were computed exclusively on the held-out test set (fixed_indices_eval_dataloader), ensuring no overlap with the training views.

To quantify visual quality, we adopted three standard novel view synthesis metrics: (i) **PSNR** (Peak Signal-to-Noise Ratio), to measure pixel-wise fidelity; (ii) **SSIM** (Structural Similarity Index Measure) [10], to evaluate structural coherence; and (iii) **LPIPS** (Learned Perceptual Image Patch Similarity) [11]. Following the protocol established by Niemeyer et al. [16], LPIPS was computed using Visual Geometry Group (VGG) features to accurately quantify perceptual distance.

Operational efficiency was measured through the inference rendering rate in frames per second (fps) across the test cameras. For memory evaluation, we analyzed both the static parameter footprint of the models and the dynamic VRAM allocation peak logged by PyTorch during runtime.

To guarantee environmental isolation and reproducibility, all experiments were executed within Nvidia Docker containers, preventing external processes from adding noise to the performance metrics. The hardware setup consisted of an NVIDIA GeForce GTX 1660 Super (6GB VRAM) running CUDA 12.6.

III. RESULTS AND DISCUSSION

A. Quantitative Analysis

B. Qualitative Analysis

C. Computational Trade-offs

IV. CONCLUSION

ACKNOWLEDGMENT

The authors would like to thank...

REFERENCES

- [1] B. Kerbl, G. Kopanas, T. Leimkühler, and G. Drettakis, “3d gaussian splatting for real-time radiance field rendering,” *ACM Transactions on Graphics*, vol. 42, no. 4, July 2023. [Online]. Available: <https://repo-sam.inria.fr/fungraph/3d-gaussian-splatting/>
- [2] M. Tancik, E. Weber, E. Ng, R. Li, B. Yi, T. Wang, A. Kristoffersen, J. Austin, K. Salahi, A. Ahuja, D. McAllister, J. Kerr, and A. Kanazawa, “Nerfstudio: A modular framework for neural radiance field development,” in *Special Interest Group on Computer Graphics and Interactive Techniques Conference Proceedings*, ser. SIGGRAPH ’23. ACM, Jul. 2023, p. 1–12. [Online]. Available: <http://dx.doi.org/10.1145/3588432.3591516>
- [3] Adobe, *Substance 3D Viewer (beta)*, 2025, computer Software. [Online]. Available: <https://helpx.adobe.com/substance-3d-viewer/get-started/quick-start.html>
- [4] T. Müller, A. Evans, C. Schied, and A. Keller, “Instant neural graphics primitives with a multiresolution hash encoding,” *ACM Transactions on Graphics*, vol. 41, no. 4, p. 1–15, Jul. 2022. [Online]. Available: <http://dx.doi.org/10.1145/3528223.3530127>
- [5] A. Yu, V. Ye, M. Tancik, and A. Kanazawa, “pixelnerf: Neural radiance fields from one or few images,” 2021. [Online]. Available: <https://arxiv.org/abs/2012.02190>
- [6] Z. Wang, M. Varma, P. Wang, X. Chen, T. Chen, and S. Venugopalan, “Is attention all that nerf needs?” 2023. [Online]. Available: <https://arxiv.org/abs/2207.13298>
- [7] K. Schwarz, Y. Liao, M. Niemeyer, and A. Geiger, “Graf: Generative radiance fields for 3d-aware image synthesis,” 2021. [Online]. Available: <https://arxiv.org/abs/2007.02442>
- [8] H. Wu, B. Xiao, N. Codella, M. Liu, X. Dai, L. Yuan, and L. Zhang, “Cvt: Introducing convolutions to vision transformers,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, October 2021, pp. 22–31.
- [9] P. Hedman, P. P. Srinivasan, B. Mildenhall, J. T. Barron, and P. Debevec, “Baking neural radiance fields for real-time view synthesis,” 2021. [Online]. Available: <https://arxiv.org/abs/2103.14645>
- [10] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, “Image quality assessment: from error visibility to structural similarity,” *IEEE transactions on image processing*, vol. 13, no. 4, pp. 600–612, 2004.
- [11] R. Zhang, P. Isola, A. A. Efros, E. Shechtman, and O. Wang, “The unreasonable effectiveness of deep features as a perceptual metric,” in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2018, pp. 586–595.
- [12] M. Niemeyer, J. T. Barron, B. Mildenhall, M. S. Sajjadi, A. Geiger, and N. Radwan, “Regnerf: Regularizing neural radiance fields for view synthesis from sparse inputs,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 5480–5490.
- [13] H. Heo, T. Kim, J. Lee, J. Lee, S. Kim, H. Kim, and J.-H. Kim, “Robust camera pose refinement for multi-resolution hash encoding,” 02 2023.
- [14] J. Kulhanek and T. Sattler, “Nerfbaselines: Consistent and reproducible evaluation of novel view synthesis methods,” 2025. [Online]. Available: <https://arxiv.org/abs/2406.17345>
- [15] B. Mildenhall, P. P. Srinivasan, R. Ortiz-Cayon, N. K. Kalantari, R. Ramamoorthi, R. Ng, and A. Kar, “Local light field fusion: Practical view synthesis with prescriptive sampling guidelines,” 2019. [Online]. Available: <https://arxiv.org/abs/1905.00889>
- [16] M. Niemeyer, J. T. Barron, B. Mildenhall, M. S. M. Sajjadi, A. Geiger, and N. Radwan, “Regnerf: Regularizing neural radiance fields for view synthesis from sparse inputs,” 2021. [Online]. Available: <https://arxiv.org/abs/2112.00724>
- [17] T. Müller, “tiny-cuda-nn,” <https://github.com/NVlabs/tiny-cuda-nn>, 2021, version 1.7, BSD-3-Clause License.